LINEAR QUEUE USING ARRAY - C PROGRAM

1. PROGRAM CODE

```c
#include <stdio.h>
#define SIZE 5
int queue[SIZE];
int front = -1, rear = -1;
void enqueue()
{
    int value;
    if (rear == SIZE - 1)
    {
        printf("Queue Overflow\n");
        return;
    }
    printf("Enter element: ");
    scanf("%d", &value);
    if (front == -1)
        front = 0;

    rear++;
    queue[rear] = value;

    printf("%d inserted into queue\n", value);
}
void dequeue()
{
    if (front == -1 || front > rear)
        printf("Queue Underflow\n");
    else
    {
        printf("%d deleted from queue\n", queue[front]);
        front++;
        if (front > rear)
        {
            front = rear = -1;
        }
    }
}
void display()
{
    int i;
    if (front == -1 || front > rear)
    {
        printf("Queue is empty\n");
    }
    else
    {
        printf("Queue: ");
        for (i = front; i <= rear; i++)
        {
            printf("%d ", queue[i]);
        }
        printf("\n");
    }
}
int main()
{
    int choice;
    while (1)
    {
        printf("\n1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1:
                enqueue();
                break;

            case 2:
                dequeue();
                break;

            case 3:
                display();
                break;

            case 4:
                printf("Exiting...\n");
                return 0;

            default:
                printf("Invalid choice\n");
        }
    }
    return 0;
}
```

2. ACTUAL PROGRAM OUTPUT

Representative full run (captured stdout)

```
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: Enter element: 10 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: Enter element: 20 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: Queue: 10 20

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 10 deleted from queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: Queue: 20

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: Exiting...
```